

Matthew do Santos-Zounon

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Richmond Hill, ON

EXPERIENCE

Software Integration Engineer

Sep 2024 – Apr 2025

Rivian — Rivian and Volkswagen Group Technologies

Vancouver, BC

- Led sub-team of 4 to ship mobile release v2.21, aligning PM, Design, & Devs through complete release cycle
- Developed automated UI and API tests using **Python's PyTest** and **Appium** integrated with **GitHub Actions CI/CD** to run on simulated and test-bench vehicles
- Developed tests for Android using **Android Studio SDK (Java & Kotlin)** and for iOS using **XCode (Swift)**, integrated these tests with vehicle hardware **HIL** and emulated vehicles **SIL**
- Mentored junior interns teaching test processes, development skills, and automation fundamentals

Software Engineer

May 2023 – Dec 2023

Operto Guest Technologies

Vancouver, BC

- Developed and deployed 200 E2E UI test case suite in **TypeScript** using **Playwright**, covering new features and 85% of the most common user flows identified through **FullStory**
- Enabled fast feature iteration by incorporating automated test suite into CI/CD pipeline
- Led cross-team bug triage aligning on severity and ownership, removing blockers for more efficient feature release

Software Systems Engineer

May 2022 – Dec 2022

Gatekeeper Systems Inc.

Abbotsford, BC

- Automated 150 manual test cases by reverse engineering **HTTPS** protocol from PCAPs (**Wireshark**, **tcpdump**) and developing a Python **scapy** tool to emulate device traffic, eliminating the need for physical hardware
- Performed static and dynamic firmware analysis on embedded **MIPS** Linux by extracting images with **binwalk**, analyzing with **radare2**, **Ghidra**, and demonstrated PoC user-space emulation with **QEMU**
- Developed concurrent network and log analysis tool that reports anomalous device behavior; now standard in QA toolkit

EXTRA-CURRICULAR

Software Team Lead

Jan 2023 – Sep 2024

SFU Autonomous Robot Soccer Club

Burnaby, BC

- Led software team in design and development of [autonomous robot software](#) in **C++** using **Qt6** and **Protobuf**
- Hosted weekly software team meetings and collaborated across sub-teams in weekly admin meetings
- Owned recruitment by reviewing applications, conducting technical interviews, and onboarding new members
- Collaborated with electrical, firmware, and mechanical teams to build robots for [RoboCup](#)

CURRICULAR PROJECTS

Parabix ARM | C++, SIMD, ARM NEON & SVE intrinsics

Compiler Technology - CMPT 980

- Implemented 67% improvement in performance of Parabix (**parallel computing**) on **AArch64** CPUs using **ARM NEON** and **SVE ISAs**
- Proved performance in **QEMU** emulated environment and confirmed results on Raspberry Pi hardware
- [Technical Talk Slides](#) | [GitLab Code PR](#)

EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Graduated Sep 2025